

Numerical Differentiation of Elliptical Planetary Orbits

A comparison of the eight planets in the Solar System

Abstract

This project uses numerical differentiation to recover orbital velocity and acceleration from discrete position data for Mercury, Venus, Earth, Mars, Jupiter, Saturn, Uranus, and Neptune. All elliptical positions were generated by solving Kepler's equation at equally spaced times. Forward, backward, and central first differences were compared with analytical velocities, and a three-point central second difference was compared with Newtonian acceleration. Mercury's large eccentricity produced the strongest variation over an orbit, so it was used for the convergence study. Halving the time step reduced forward- and backward-speed errors by about a factor of two and central-speed and acceleration errors by a factor of four. The results confirmed first-order convergence for the one-sided formulas and second-order convergence for the centered formulas.

1 Introduction

Planetary motion is a natural application of numerical differentiation because observations and simulations typically provide positions at discrete times rather than continuous functions. Velocity must then be estimated from the first derivative of position, and acceleration from the second derivative. These quantities reveal both the changing motion along an elliptical orbit and the inward gravitational force that sustains it.

The eight planets span a wide range of physical characteristics. Mercury completes an orbit in about 88 days and has an eccentricity near 0.206, whereas Neptune requires roughly 59,800 days and follows a nearly circular path with eccentricity near 0.010 [1]. This range makes the Solar System a useful test of whether the same finite-difference algorithms behave consistently across different spatial scales.

The objective was to generate discrete heliocentric position data for all eight planets, compute velocity using forward, backward, and central differences, compute acceleration using a three-point second derivative, and compare the numerical results with exact two-body values. A second objective was to measure convergence as the time step decreased and to identify the most accurate method.

2 Theory

2.1 Elliptical Orbit Model

Each planet was modeled as a point mass moving along a fixed, planar ellipse with the Sun at one focus. For orbital period T and time t , the mean anomaly M increases uniformly. The eccentric anomaly E was obtained from Kepler's equation:

$$M = \frac{2\pi t}{T}, M = E - e \sin E$$

Because E appears both inside and outside the sine function, Kepler's equation was solved iteratively with Newton's method. The planar coordinates were then computed from the semimajor axis a and the eccentricity e :

$$x = a(\cos E - e), y = a\sqrt{1 - e^2} \sin E$$

These are the same orbital-plane relations JPL uses to compute approximate planetary positions [2]. The radius varies from $a(1-e)$ at perihelion to $a(1+e)$ at aphelion. Equal time increments therefore do not produce equal spatial increments: a planet moves farther between samples near perihelion than near aphelion.

2.2 Numerical Differentiation

For uniformly spaced data with time step Δt , the forward and backward formulas estimate the derivative from adjacent samples:

$$f'_F(t_i) \approx \frac{f_{i+1} - f_i}{\Delta t}, f'_B(t_i) \approx \frac{f_i - f_{i-1}}{\Delta t}$$

Both one-sided formulas have leading truncation error proportional to Δt . The central first difference uses samples on both sides of the evaluation point, and the three-point second difference estimates acceleration directly from position:

$$f'_C(t_i) \approx \frac{f_{i+1} - f_{i-1}}{2\Delta t}, f''_F(t_i) \approx \frac{f_{i+1} - 2f_i + f_{i-1}}{(\Delta t)^2}$$

The centered formulas have a leading truncation error proportional to $(\Delta t)^2$ for a smooth function [3]. The orbital arrays were treated as periodic: the final sample connects to the first sample one orbital period later. This provides valid neighbors at both ends without discarding data.

2.3 Analytical Comparison

Analytical velocity components were obtained by differentiating the eccentric-anomaly coordinates. With mean motion $2\pi/T$, the required derivatives are:

$$\frac{dE}{dt} = \frac{2\pi/T}{1 - e \cos E}, v_x = -a \sin E \frac{dE}{dt}, v_y = a \sqrt{1 - e^2} \cos E \frac{dE}{dt}$$

The exact two-body acceleration points toward the Sun. Kepler's third law was used to express the gravitational parameter in terms of each tabulated period and semimajor axis:

$$a = \frac{\mu}{r^3}, \mu = \left(\frac{2\pi}{T}\right)^2 a^3$$

Numerical errors were calculated point by point and summarized with the mean absolute error (MAE):

$$\varepsilon_i = |f_{\text{numerical},i} - f_{\text{exact},i}|, MAE = \frac{1}{N} \sum_{i=1}^N \varepsilon_i$$

3 Computational Method

The Python program used NumPy for arrays and Matplotlib for plots. For each planet, 400 equally spaced times were generated over one complete period. Newton's method was applied for ten iterations at each time to solve Kepler's equation. The resulting x and y coordinates were differentiated separately, then combined to obtain the magnitudes of

speed and acceleration. Velocities were converted from AU/day to km/s, and accelerations from AU/day² to m/s².

A separate `numerical_methods.py` module contains the four required finite-difference algorithms. The main program stores numerical and exact quantities, writes summary and convergence CSV files, and generates five figures. The orbital inputs are rounded values consistent with NASA planetary data [1].

Table 1. Orbital parameters used by the model

Planet	a (AU)	e	Period (days)
Mercury	0.387	0.2056	87.97
Venus	0.723	0.0068	224.70
Earth	1.000	0.0167	365.25
Mars	1.524	0.0934	686.98
Jupiter	5.203	0.0489	4,332.59
Saturn	9.537	0.0565	10,759.22
Uranus	19.191	0.0472	30,688.50
Neptune	30.069	0.0086	60,182.00

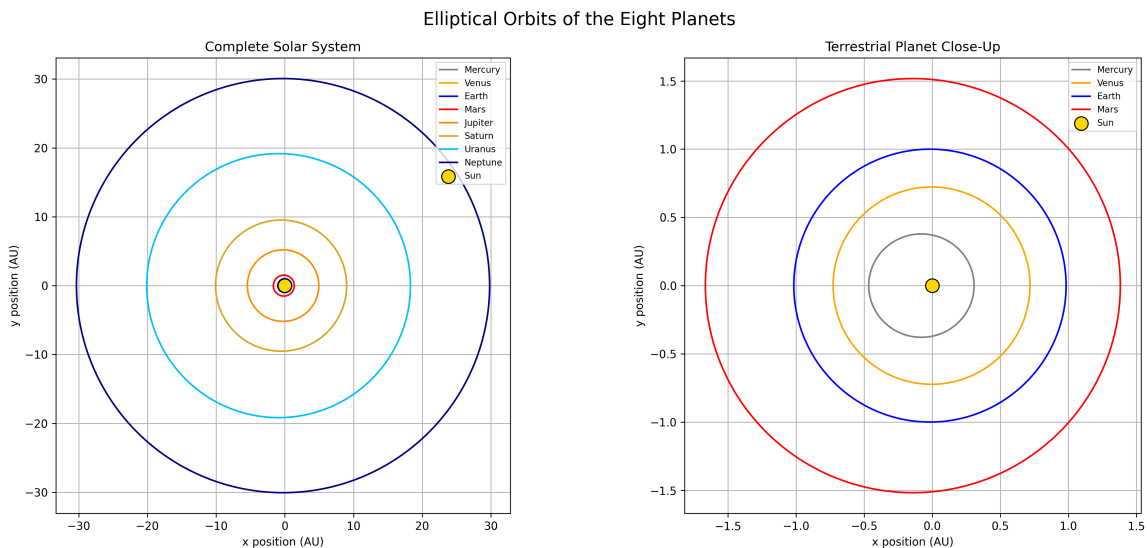


Figure 1. Smooth elliptical paths for all eight planets and a close-up of the terrestrial planets.

4 Results

The orbit plots capture the large-scale difference between the inner and outer Solar System. A separate terrestrial-planet panel is necessary because Neptune's semimajor

axis is about 30 AU, which compresses the inner four orbits on a single scale. The equal-time markers in Figure 2 make the changing speed visible. Mercury's markers are widely spaced near perihelion and crowded near aphelion, whereas Venus and Neptune show nearly uniform spacing because their eccentricities are small.

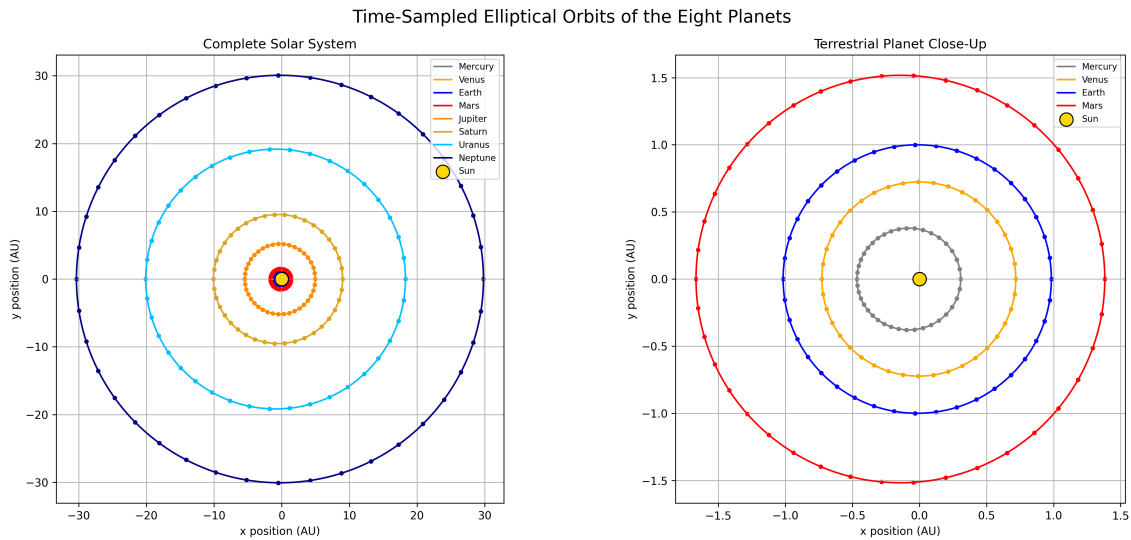


Figure 2. Equal-time samples along the modeled orbits. Marker spacing reveals changes in orbital speed.

Table 2. Selected exact orbital results

Planet	Max speed (km/s)	Min speed (km/s)	Max accel. (m/s ²)
Mercury	58.959	38.850	6.269e-02
Venus	35.244	34.768	1.148e-02
Earth	30.287	29.292	6.133e-03
Mars	26.504	21.976	3.108e-03
Jupiter	13.720	12.441	2.424e-04
Saturn	10.204	9.113	7.322e-05
Uranus	7.132	6.489	1.776e-05
Neptune	5.483	5.389	6.683e-06

4.1 Velocity and acceleration

The greatest variation in velocity occurs for Mercury. Its modeled speed ranges from 38.85 km/s near aphelion to 58.96 km/s near perihelion. Earth's smaller eccentricity yields a narrower range of 29.29 to 30.29 km/s. Neptune is slowest, remaining between 5.39 and 5.48 km/s. These results align with the expected trend: planets closer to the Sun move faster, and eccentric orbits exhibit larger within-orbit variations.

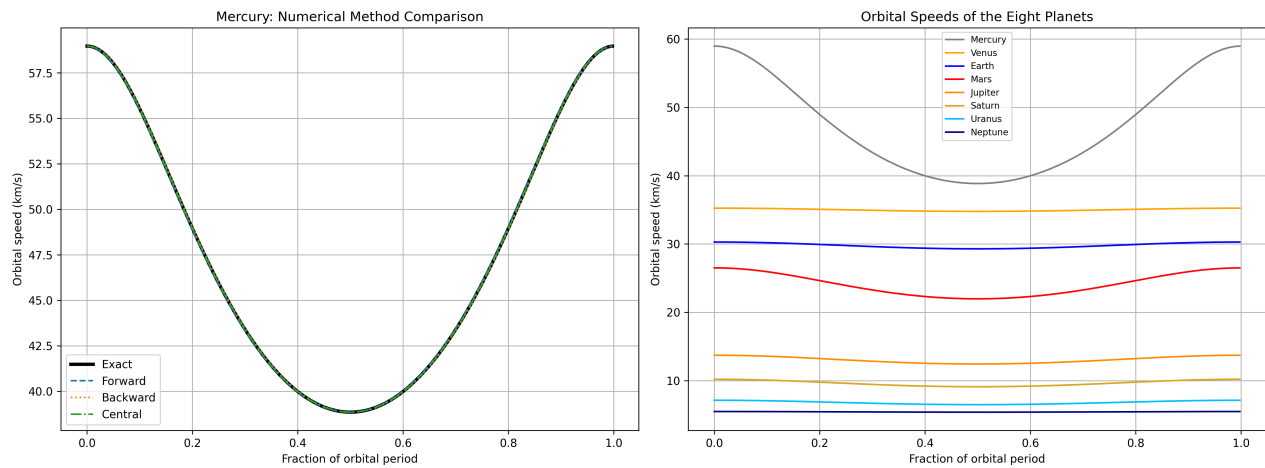


Figure 3. Mercury velocity-method comparison and exact orbital speeds for all eight planets.

At 400 samples per orbit, the forward- and backward-Mercury speed MAEs are both 5.03×10^{-2} km/s, whereas the central-difference MAE is 2.08×10^{-3} km/s. The one-sided methods yield nearly identical mean errors because they sample the same periodic curve in opposite directions. The numerical curves visually overlap the analytical curve at this plotting scale, so the error plot in Section 5 is needed to distinguish their accuracy.

4.2 Acceleration

Acceleration decreases rapidly with orbital distance. Mercury reaches $6.27 \times 10^{-2} \text{ m/s}^2$ near perihelion, whereas Neptune reaches only $6.68 \times 10^{-6} \text{ m/s}^2$. A logarithmic vertical axis is therefore used in the all-planet comparison. The three-point numerical acceleration matches the exact inward gravitational acceleration throughout Mercury's orbit.

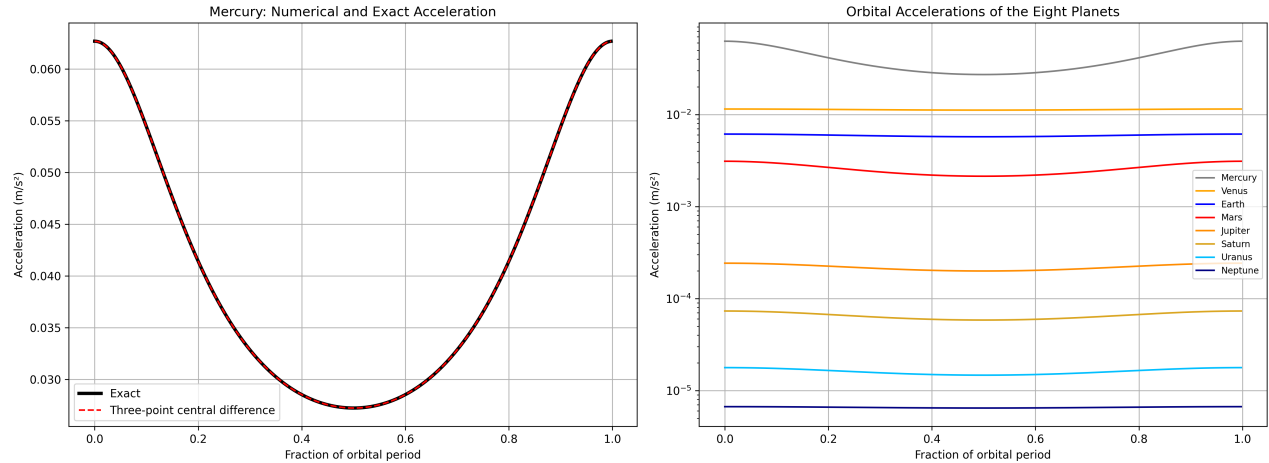


Figure 4. Numerical and exact acceleration for Mercury, with exact acceleration curves for all planets.

5 Error Analysis

Mercury was evaluated using 50, 100, 200, 400, 800, and 1600 points per orbit. The corresponding time step decreased from 1.7594 days to 0.05498 days. Halving the step reduced the forward- and backward-speed errors by almost exactly a factor of two, demonstrating first-order convergence. The central speed error and the second-derivative acceleration error decreased by almost exactly a factor of four, demonstrating second-order convergence.

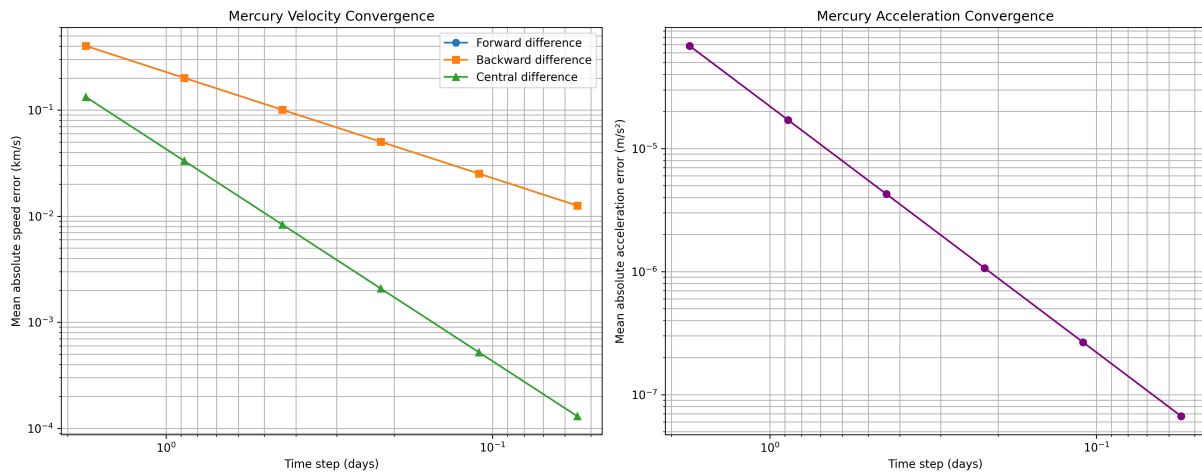


Figure 5. Mean absolute velocity and acceleration errors as Mercury's time step decreases.

Table 3. Mercury mean absolute errors

N	dt (days)	Forward v	Backward v	Central v	Central a
50	1.75940	4.02e-01	4.02e-01	1.33e-01	6.82e-05
100	0.87970	2.01e-01	2.01e-01	3.33e-02	1.71e-05

N	dt (days)	Forward v	Backward v	Central v	Central a
200	0.43985	1.01e-01	1.01e-01	8.32e-03	4.27e-06
400	0.21993	5.03e-02	5.03e-02	2.08e-03	1.07e-06
800	0.10996	2.51e-02	2.51e-02	5.20e-04	2.67e-07
1600	0.05498	1.26e-02	1.26e-02	1.30e-04	6.67e-08

5.1 Sources of numerical error

Truncation error is the dominant numerical error over the tested range. A finite-difference formula approximates a derivative using a truncated Taylor series, so reducing Delta t improves the result at the expected rate. At extremely small steps, roundoff error would eventually grow because nearly equal positions are subtracted and divided by a small number; that regime was not reached in this experiment.

The calculation also includes model error distinct from numerical error. Each orbit is a fixed planar two-body ellipse with the Sun at one focus. The model omits orbital inclination, precession, interactions among planets, the Sun's barycentric motion, and relativistic corrections. The orbital elements are rounded and treated as constant. JPL notes that Keplerian approximations are lower-accuracy alternatives to integrated ephemerides such as Horizons [2]. These limitations affect physical realism but do not invalidate the comparison of differentiation formulas on the chosen smooth data.

Forward and backward errors are equal in the Mercury MAE table because the periodic sampling is uniform and the two formulas traverse the same closed trajectory in opposite directions. Central differencing is the best-performing method asymptotically. For a nearly circular orbit, however, cancellation in the scalar speed magnitude can occasionally make a one-sided speed error appear competitive at a fixed resolution, even though the central component derivatives retain the higher formal order.

6 Conclusions

Numerical differentiation successfully recovered the orbital velocity and acceleration of all eight planets from discrete position data. The results were consistent with Keplerian motion: inner planets moved faster and accelerated more strongly, and all planets moved fastest at perihelion. Mercury exhibited the largest speed change because it had the greatest eccentricity among the modeled planets.

The convergence experiment identified central differencing as the preferred method. Forward and backward differences converged linearly with step size, whereas the central first difference and the three-point second difference converged quadratically. At 400 Mercury samples, the central speed MAE was about twenty-four times smaller than either

one-sided speed MAE. Future improvements could include JPL Horizons ephemerides, three-dimensional orbital orientations, perturbations from other planets, adaptive time sampling near perihelion, and a study of measurement noise.

References

- [1] NASA Space Science Data Coordinated Archive. Planetary Fact Sheet - Metric. Updated March 18, 2025. <https://nssdc.gsfc.nasa.gov/planetary/factsheet/>
- [2] Jet Propulsion Laboratory Solar System Dynamics. Approximate Positions of the Planets. https://ssd.jpl.nasa.gov/planets/approx_pos.html
- [3] R. L. Burden, J. D. Faires, and A. M. Burden. Numerical Analysis. 10th ed., Cengage Learning, 2016.